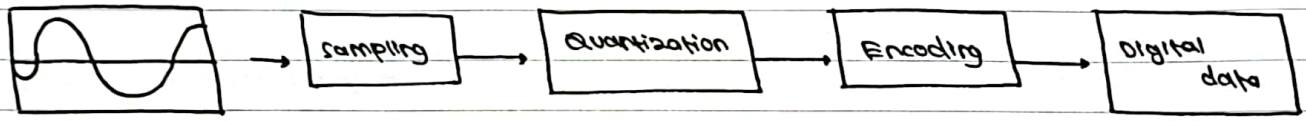
— Digital Transmisi —

No 01-September-2023
Date Jumat

- Alasan sinyal analog perlu dirubah ke sinyal digital :
  1. Tahan terhadap noise
  2. Dapat (mudah dipulihkan, diteteksi dan dipetuk
- contoh : perubahan TV analog ke TV digital
- PCM merupakan code modulus pulsa yang berfungsi
- terdapat 3 step untuk mendigitalkan sinyal analog :
  1. Sampling
     ↳ sebelum tahap ini perlu adanya filter untuk menghilangkan noise ke minimum
     ↳ harus mempertahankan frekuensi tinggiya, karena dapat merubah data.
  2. Quantization
  3. Binary Encoding

- Prosses :



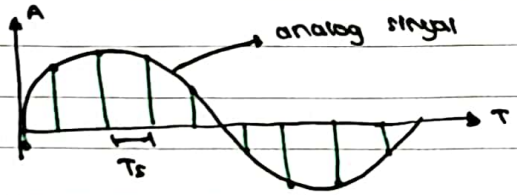
analog sinyal

SAMPLING

- sinyal analog diambil sampelnya setiap Ts secs.
- Ts disebut sebagai interval pengambilan sampel (sampling interval)
- $Fs = \frac{1}{Ts}$ disebut laju pengambilan sampel / frekuensi pengambilan sampel (sampling rate / sampling frekuensi).
- terdapat 3 metode pengambilan sampel :

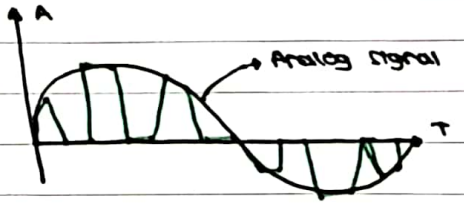
①. Ideal sampling

↳ impuls pada setiap pengambilan sampel



②. Natural sampling

↳ pulsa dengan lebar persis dengan amplitudo berubah

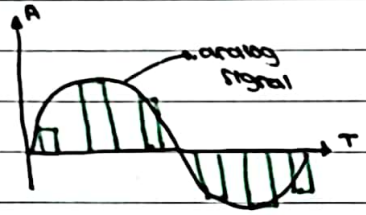


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③. Flattop sampling

↳ Sampel dan hold seperti natural tetapi dengan nilai amplitudo tunggal



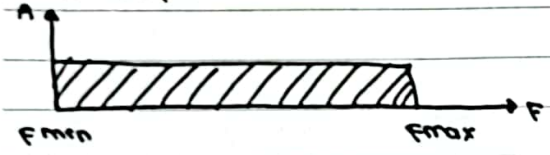
- proses ini disebut sebagai modulasi amplitudo pulsa (PAM) dan keluaran berupa sinyal dengan nilai analog (non integer).

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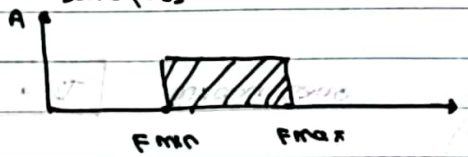
No
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- Amplitudo flat top akan sama
- prosesnya tergantung pada PAM
- teori Nyquist : " bahkan akan menyimpang frekuensi sinyalnya harus 2x frekuensi sinyal " untuk bisa memenuhi proses sinyal
- Nyquist teorema pada bandpass = low pass

• Lowpass



• Bandpass

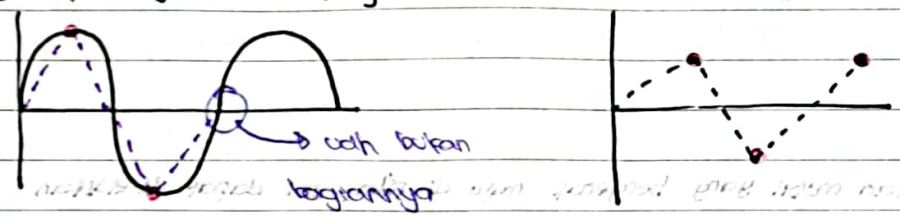


Nyquist rate : 2 x Fmax

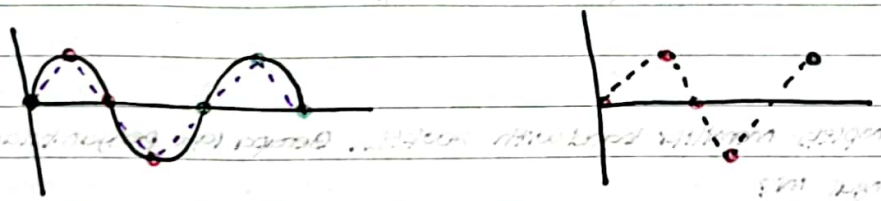
- pada soal diperlukan : panjang frekuensi dan nilai Fmax / Fmin
- Alasan perlu diketahui panjang dan frekuensi agar memenuhi nilai sampling nya.

** contoh :

Fs = 2F = nyquist sampling

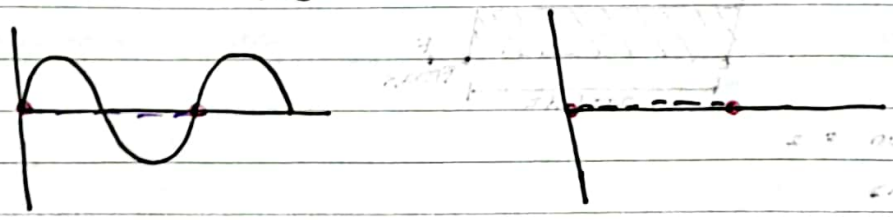


Fs = 4F = oversampling



tak direkomendasikan,
karena buang2

Fs = 1F = undersampling

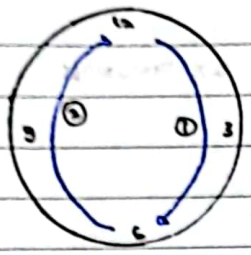


tak direkomendasikan,
karena sampling tak terjadi

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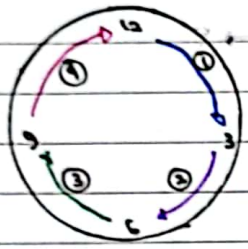
contoh :



Nyquist rate $T_s = T\frac{1}{2}$

pola : 12 - 6 - 12 - 6 - 12

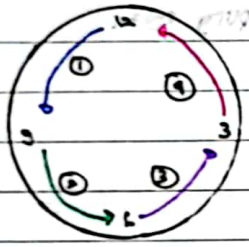
Dapat berubah arah



oversampling $T_s = T\frac{1}{4}$

pola : 12 - 3 - 6 - 9 - 12

pola hanya maju



undersampling $T_s = T\frac{3}{4}$

pola : 12 - 9 - 6 - 3 - 12

- contoh yang nyata : Gerakan mobil yang bergerak maju di film, ini dapat dijelaskan dengan undersampling. Jika roda sedang berputar 12 r/s
- perusahaan telepon mendigitalkan suara dengan asumsi frekuensi maksimum 9000 Hz. oleh karena itu, kecepatan pengambilan sampel adalah 8000 /s

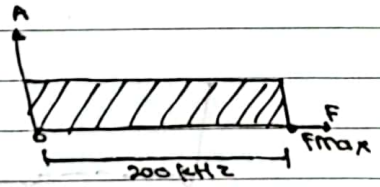
Latihan soal

1. Sinyal low pass yang komplek memiliki bandwith 200kHz. Berapa laju pengambilan sample minimum untuk sinyal ini?

Jawab :

Bw = 200 kHz

Lowpass : 0 — P



maka : $f_s = 2.F_{max}$

$= 2.\ 200\ kHz$

$F_s = 400\ kHz$ //

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2. Sinyal bandpass komplets memiliki bandwith 200kHz. Berapa laju pengambilan sampel minimum untuk sinyal ini?

Jawab :

Bw = 200 kHz

bandpass : diperlukan diketahui F_{min} ataupun F_{max}

maka nilai Fs tidak dapat diketahui.

QUANTIZATION

- langkah - langkah :

1. Mencari L = banyaknya level
2. mencari $\Delta = \frac{(max - min)}{L}$
3. Cari mid - points untuk setiap zona, yaitu dari 0 sehingga $\boxed{L-1}$
4. Tiap sampling akan masuk ke salah satu zona
5. Mencari rata-rata (normalisasi) ke nilai mid-point (nilai tengah tiap level).

- Quantization level
  - titik tengah setiap zona diberi nilai dari 0 hingga L-1 (menghasilkan nilai L)
  - setiap sampel yang jatuh pada suatu zona kemudian didefasi dengan nilai titik tengahnya.

- Quantization zones
  - contoh soal :

diketahui : $V_{min} = -20V$ $L = 8$

$V_{max} = 20V$

ditanyakan : lebar zona (Δ), batas zona, titik tengahnya, kode biner, jumlah bit, dan kode ke zona ?!

Jawab :

lebar zona $\Delta = \frac{(max - min)}{L} = \frac{20 - (-20)}{8} = \frac{40}{8} = 5$ zona

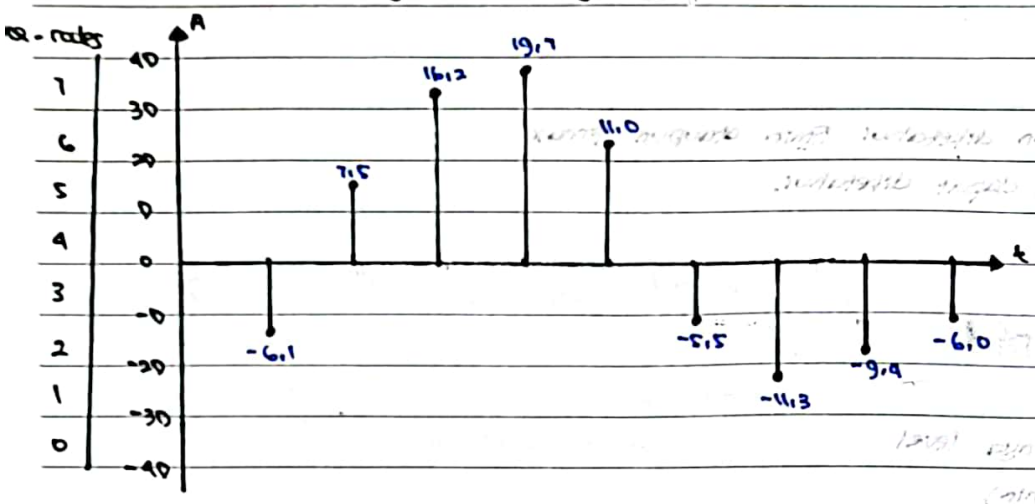
batas zona / definisi :	-20 to -15	0 to +5
tiap - tiap level	-15 to -10	+5 to +10
	-10 to -5	+10 to +15
	-5 to 0	+15 to +20

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Nilai titik tengah : -17,5 : -12,5 : -7,5 : -2,5 : 2,5 : 7,5 : 12,5 : 17,5

kuantisasi dan pengkodean sinyal sampel :



Normalized PAM values	-1,22	-1,50	3,29	3,99	2,20	-1,10	-2,26	-1,08	-1,20
Normalized Q-values	-1,50	1,50	3,50	3,50	2,50	-1,50	-2,50	-1,50	-1,50
Normalized error	-0,30	0	+0,26	-0,99	+0,30	-0,40	-0,29	+0,38	-0,20
Q-code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

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KONVOLUSI

↳ proses pengolahan 2 sinyal

- semua yang berhubungan dengan konvolusi disebut filter || fungsi : filter

- persamaan :

$$y[m] = x[n] \circledast h[k]$$

Implus respon

$x[n]$ → $h[k]$ → $y(m)$

input proses output

- konvolusi dibagi menjadi 2 :

① konvolusi secara matematis

$$Y[m] = \sum_{n=0}^{N-1} x[n] \cdot h[m-n]$$

$$M = N + k - 1$$

- contoh :

Diketahui $x[n] = [\overset{0}{6}\ \overset{1}{4}\ \overset{2}{3}]$: $h[k] = [\overset{0}{1}\ \overset{1}{3}\ \overset{2}{4}\ \overset{3}{5}]$

Jawab :

$M = N + k - 1$

$= 3 + 4 - 1$

$M = 6$ → maka terdapat 6 buah proses

— proses 1 —

$Y[0] = x[0] . h[0-0] + x[1] . h[0-1]$

$= x[0] . h[0]$

$= 6 . 1$

$Y[0] = 6$

— proses 2 —

$Y[1] = x[0] . h[1-0] + x[1] . h[1-1] + x[2] . h[1-2]$

$= x[0] . h[1] + x[1] . h[0]$

$= (6 . 3) + (4 . 1)$

$= 18 + 4$

$Y[1] = 22$

— proses 3 —

$Y[2] = x[0] . h[2-0] + x[1] . h[2-1] + x[2] . h[2-2] + x[3] . h[2-3]$

$= x[0] . h[2] + x[1] . h[1] + x[2] . h[0]$

$= (6 . 4) + (4 . 3) + (3 . 1)$

Dipindai dengan CamScanner

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= 24 + 12 + 3

= 39 //

— proses 4 —

$y(3) = x[0].h[3-0] + x[1].h[3-1] + x[2].h[3-2] + x[3].h[3-3] + x[4].h[3-4]$

$= x[0].h[3] + x[1].h[2] + x[2].h[1]$

= (6.5) + (4.4) + (3.3)

= 30 + 16 + 9 + 0

= 55 //

— proses 5 —

$y[4] = x[0].h[4-0] + x[1].h[4-1] + x[2].h[4-2] + x[3].h[4-3]$

$= x[1].h[3] + x[2].h[2]$

= (4.5) + (3.4)

= 20 + 12

= 32 //

— proses 6 —

$y[5] = x[0].h[5-0] + x[1].h[5-1] + x[2].h[5-2] + x[3].h[5-3]$

$= x[2].h[3]$

= 3.5

= 15 //

2. menggunakan rumus berbeda

$$y[m] = \sum_{k=0}^{k-1} h[k].x[m-k]$$

Jawab :

— proses 1 —

$y[0] = h[0].x[0-0] + h[1].x[0-1]$

$= h[0].x[0]$

= 1.6

= 6 //

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— proses 2 —

$y[1] = h[0] \cdot x[1-0] + h[1] \cdot x[1-1] + h[2] \cdot x[1-2]$

$= h[0] \cdot x[1] + h[1] \cdot x[0]$

$= (1.4) + (3.6)$

$= 4 + 18$

$= 22$

— proses 3 —

$y[2] = h[0] \cdot x[2-0] + h[1] \cdot x[2-1] + h[2] \cdot x[2-2] + h[3] \cdot x[2-3]$

$= h[0] \cdot x[2] + h[1] \cdot x[1] + h[2] \cdot x[0]$

$= (1.3) + (3.4) + (4.6)$

$= 3 + 12 + 24$

$= 39$

— proses 4 —

$y[3] = h[0] \cdot x[3-0] + h[1] \cdot x[3-1] + h[2] \cdot x[3-2] + h[3] \cdot x[3-3]$

$= h[1] \cdot x[2] + h[2] \cdot x[1] + h[3] \cdot x[0]$

$= (3.3) + (4.4) + (5.6)$

$= 9 + 16 + 30$

$= 55$

— proses 5 —

$y[4] = \ldots h[2] \cdot x[4-2] + h[3] \cdot x[4-3]$

$= h[2] \cdot x[2] + h[3] \cdot x[1]$

$= (4.3) + (5.4)$

$= 12 + 20$

$= 32$

— proses 6 —

$y[5] = \ldots h[3] \cdot x[5-3]$

$= h[3] \cdot x[2]$

$= 5.3$

$= 15$

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✱ kesimpulan

$y[m] = x[n] . h[m-n]$ akan sama $y[m] = h[k] . x[m-k]$ sifat komutatif

$x . (h.g) = (x.h) . g$ sifat asosiatif

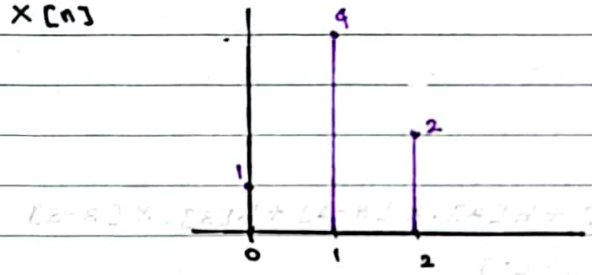
$x . (h_1 + h_2) = x.h_1 + x.h_2$ sifat distributif

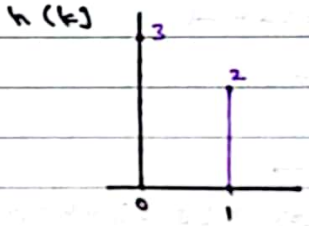
③. konvolusi secara grafik

$x[n] = [\overset{0}{1}\ \overset{1}{4}\ \overset{2}{2}]$

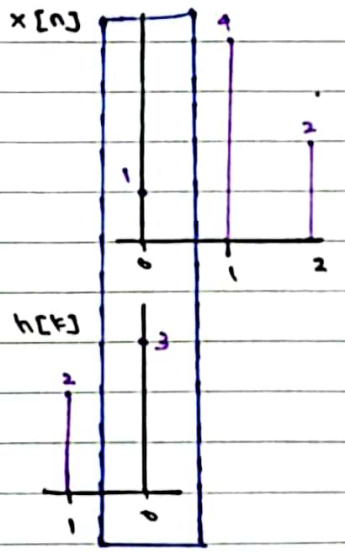
$h[k] = [\overset{0}{3}\ \overset{1}{2}]$

Jawab :

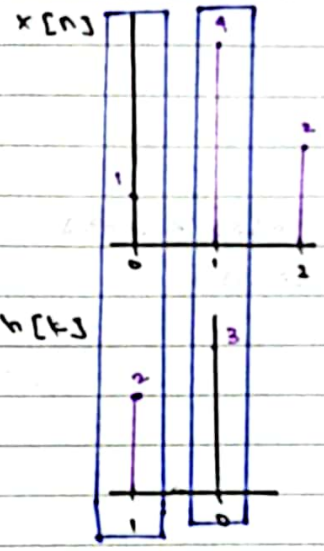




(dibuat konvetri)



maka, $y[0] = 3 \times 1 = 3$ //

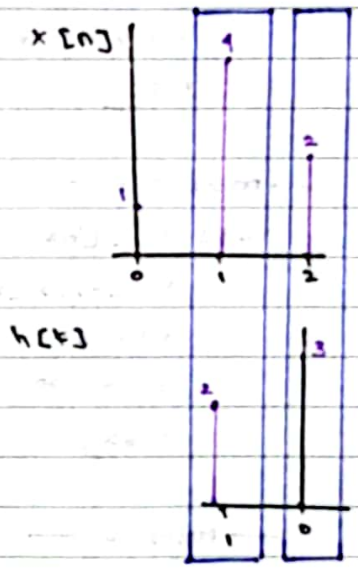


maka,

$y[1] = (2.1) + (4.3)$

$= 2 + 12$

$y[1] = 14$ //



maka,

$y[2] = (4.2) + (2.3)$

$= 8 + 6$

$y[2] = 14$

maka, $y[m] = [3\ 14\ 14]$ //

Dipindai dengan CamScanner

No tgl : 15 - sept - 2023
Date Jumat

• Jenis - jenis filter

x → [h] → y

1. Low Pass Filter (LPF)

- syarat h : - semua nilai positif (+)
  - jumlah nilainya =1 harus mendapatkan nilai $\sum H(x,y) = 1$
  - cara : dijumlahkan atau dibagi nilainya.
    contoh : h = [1 2 1] = 6 maka h = $\frac{1}{6}$ [1 2 1]
- fungsi : membuat blur dan menghilangkan noise

2. High Pass Filter (HPF)

↳ fungsi : mendeteksi tepi

- syarat h : - Terdiri dari +, - dan 0
  - jumlah nilainya = 0 harus mendapatkan nilai (x,y) = 0
  - contoh : h = [000 -1 +1 000]

3. Band Pass Filter (BP

↳ fungsi :

- syarat h : - Terdiri dari 0, + dan -
  - jumlahnya ≠ 0 harus mendapatkan hasil $\sum H(x,y) \neq 0$

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KORELASI

↳ fungsi : pattern matching

- persamaan :

$$Y = X * t$$

- korelasi dibagi menjadi 2, yaitu :

① korelasi secara matematis

$$Y[m] = \sum_{n=0}^{m+k-1} x[m-(k-1)] \cdot t[k] \qquad M = N + K - 1$$

• contoh :

1. $x = [1\ 2\ 1\ 1\ 3\ 2]$, $N = 6$

   $t = [1\ 2\ 3]$, $k = 3$

Jawab :

$M = N + K - 1$

$= 6 + 3 - 1$

$M = 8$

—— proses 1 ——

$Y[0] = x[0-(3-1)] \cdot t[0] + x[1-(3-1)] \cdot t[1] + x[2-(3-1)] \cdot t[2]$

$Y[0] = x[0] \cdot t[3]$

$= 1 \cdot 3$

$Y[0] = 3$ //

—— proses 2 ——

$Y[1] = x[1-(3-1)] \cdot t[0] + x[2-(3-1)] \cdot t[1] + x[3-(3-1)] \cdot t[2] + x[4-(3-1)] \cdot t[3]$

$= x[0] \cdot t[1] + x[1] \cdot t[2]$

$= (1 \cdot 2) + (2 \cdot 3)$

$= 2 + 6$

$Y[1] = 8$ //

—— proses 4 ——

$Y[3] = x[3-(3-1)] \cdot t[0] + x[4-(3-1)] \cdot t[1] + x[5-(3-1)] \cdot t[2]$

$= x[1] \cdot t[0] + x[2] \cdot t[1] + x[3] \cdot t[2]$

$= (2 \cdot 1) + (1 \cdot 2) + (1 \cdot 3)$

$Y[3] = 7$ //

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—— proses 3 ——

$Y[2] = x[2-(3-1)].t[0] + x[3-(3-1)].t[1] + x[4-(3-1)].t[2]$

$= x[0].t[0] + x[1].t[1] + x[2].t[2]$

$= (1.1) + (2.2) + (1.3)$

$Y[2] = 8$ //

—— proses 5 ——

$Y[4] = x[4-(3-1)].t[0] + x[5-(3-1)].t[1] + x[6-(3-1)].t[2]$

$= x[2].t[0] + x[3].t[1] + x[4].t[2]$

$= (1.1) + (1.2) + (3.3)$

$Y[4] = 12$ //

—— proses 6 ——

$Y[5] = x[5-(3-1)].t[0] + x[6-(3-1)].t[1] + x[7-(3-1)].t[2]$

$= x[3].t[0] + x[4].t[1] + x[5].t[2]$

$= (1.1) + (3.2) + (2.3)$

$Y[5] = 13$ //

—— proses 7 ——

$Y[6] = x[6-(3-1)].t[0] + x[7-(3-1)].t[1]$

$= x[4].t[0] + x[5].t[1]$

$= (3.1) + (2.2)$

$Y[6] = 7$ //

—— proses 8 ——

$Y[7] = x[7-(3-1)].t[0]$

$= x[5].t[0]$

$= 2.1$

$Y[7] = 2$ //

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Date

② . menggunakan rumus berbeda

$$Y[m] = \sum_{m=0}^{m-1} \sum_{n=0}^{N-1} t[m-(N-1)] \cdot x[n]$$

Jawab :

— proses 1 —

$Y[0] = t[5-(6-1)] \cdot x[5]$

$= t[0] \cdot x[5]$

$= (1.2)$

$= 2$ //

— proses 2 —

$Y[1] = t[5-(6-1)] \cdot x[4] + t[6-(6-1)] \cdot x[5]$

$= t[0] \cdot x[4] + t[1] \cdot x[5]$

$= (1.3) + (2.2)$

$= 7$ //

— proses 3 —

$Y[2] = t[5-(6-1)] \cdot x[3] + t[6-(6-1)] \cdot x[4] + t[7-(6-1)] \cdot x[5]$

$= t[0] \cdot x[3] + t[1] \cdot x[4] + t[2] \cdot x[5]$

$= (1.1) + (2.3) + (3.2)$

$= 13$ //

— proses 4 —

$Y[3] = t[5-(6-1)] \cdot x[2] + t[6-(6-1)] \cdot x[3] + t[7-(6-1)] \cdot x[4] + t[8-(6-1)] \cdot x[5]$

$= t[0] \cdot x[2] + t[1] \cdot x[3] + t[2] \cdot x[4] + t[3] \cdot x[5]$

$= (1.1) + (2.1) + (3.3)$

$= 12$ //

— proses 5 —

$Y[4] = t[5-(6-1)] \cdot x[1] + t[6-(6-1)] \cdot x[2] + t[7-(6-1)] \cdot x[3]$

$= t[0] \cdot x[1] + t[1] \cdot x[2] + t[2] \cdot x[3]$

$= (1.2) + (2.1) + (3.1)$

$= 7$ //

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— Proses 6 —

$y[5] = h[5-(6-1)] \cdot x[0] + h[6-(6-1)] \cdot x[1] + h[7-(6-1)] \cdot x[2]$

$= h[0] \cdot x[0] + h[1] \cdot x[1] + h[2] \cdot x[2]$

$= (1.1) + (2.2) + (3.1)$

$= 8$ //

— Proses 7 —

$y[6] = h[6-(6-1)] \cdot x[0] + h[7-(6-1)] \cdot x[1]$

$= h[1] \cdot x[0] + h[2] \cdot x[1]$

$= (2.1) + (3.2)$

$= 8$ //

— Proses 8 —

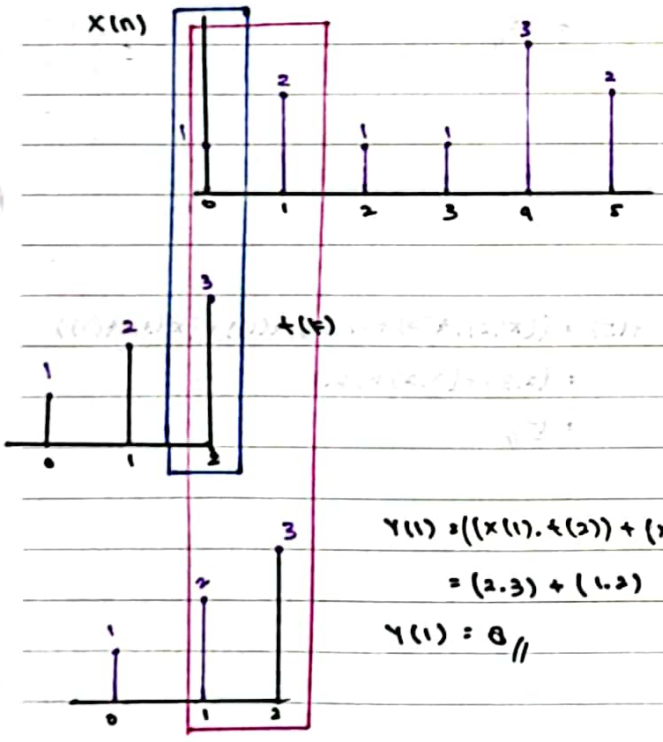
$y[7] = h[7-[6-1)] \cdot x[0]$

$= h[2] \cdot x[0]$

$= 3.1$

$= 3$ //

* menggunakan grafik



$y(0) = x(0) \cdot h(2) = 1.3 = 3$ //

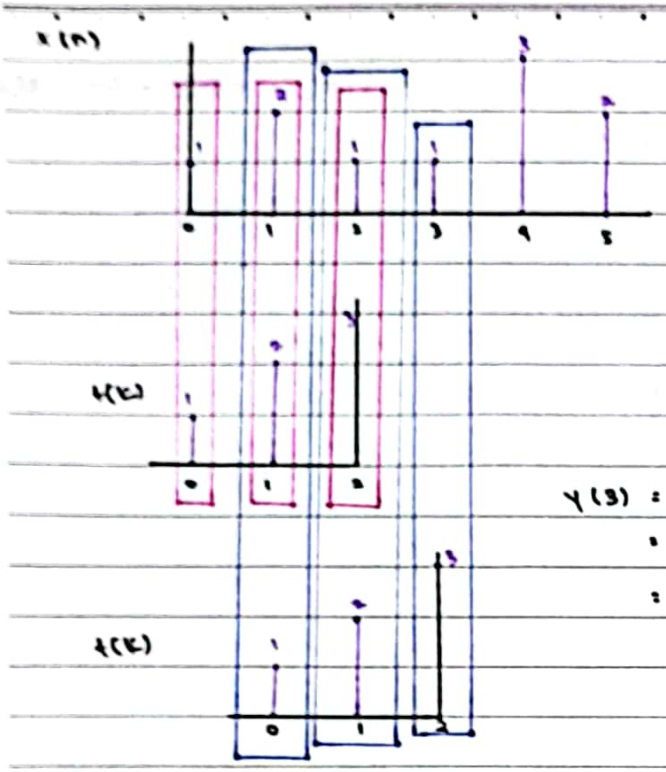
$y(1) = ((x(1) \cdot h(2)) + (x(0) \cdot h(1)))$

$= (2.3) + (1.2)$

$y(1) = 8$ //

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$Y(2) = ((x(2).f(2)) + (x(1).f(1)) + (x(0).f(0))$

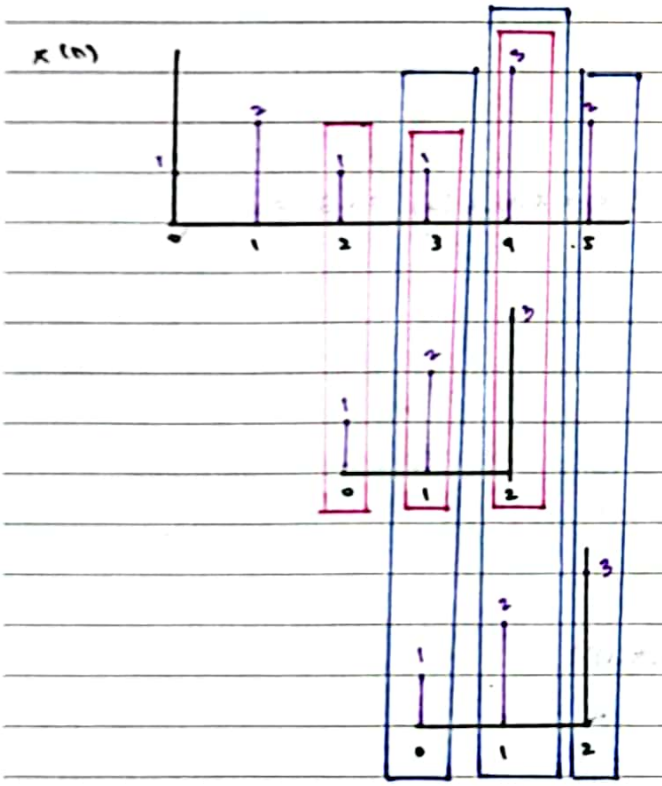
$= (1.3) + (2.2) + (1.1)$

$Y(2) = 8$ //

$Y(3) = ((x(3).f(3)) + (x(2).f(1)) + (x(1).f(0))$

$= (1.3) + (1.2) + (2.1)$

$= 7$ //



$Y(4) = ((x(4).f(2)) + (x(3).f(1)) + (x(2).f(0))$

$= (3.3) + (1.2) + (1.1)$

$= 12$ //

$Y(5) = ((x(5).f(2)) + (x(4).f(1)) + (x(3).f(0))$

$= (2.3) + (3.2) + (1.1)$

$= 13$ //

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Berdasarkan dimensinya, konvolusi dibagi menjadi :

① konvolusi 2D secara grafis

- persamaan :

$M = X + I - 1$ (kolom)

$N = Y + J - 1$ (baris)

$Y = [M, N]$

	0	1	2	3
0	10 ✓	90 ✓	70 ✓	60 ✓
1	70 ✓	230 ✓	330 ✓	240 ✓
2	120 ✓	310 ✓	380	240

M

N

- contoh soal :

$F(x,y) = [1\ 2,\ 3\ 4]$

	0	1
0	1	2
1	3	4

Y, X

Fh

2	1
4	3

FV

4	3
2	1

1 0

$I(I,J) = [10\ 20\ 30,\ 40\ 50\ 60]$

	0	1	2
0	10	20	30
1	40	50	60

J, I

$M = X + I - 1$

$= 2 + 3 - 1$

$M = 4$ kolom

$N = Y + J - 1$

$= 2 + 2 - 1$

$N = 3$ baris

Jawab :

$Y(0,0) = 1.10 = 10$ //

4	3		
2	1 10	20	30
	40	50	60

$Y(1,0) = (2.10) + (1.20) = 40$ //

4	3		
2 10	1 20	30	
40	50	60	

$Y(0,1) = (3.10) + (1.40) = 70$ //

4	3 10	20	30
2	1 40	50	60

$Y(2,0) = (2.20) + (1.30) = 70$ //

	4	3	
10	2 20	1 30	
40	50	60	

$Y = (0,2) = (3.40) = 120$ //

	10	20	30
4	3 40	50	60
2	1		

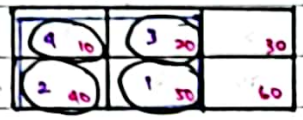
$Y(3,0) = (2.30) = 60$ //

		4	3
10	20	2 30	1
40	50	60	

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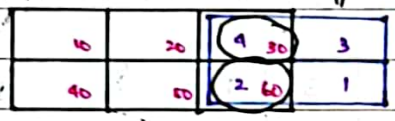
$Y(1,1) = ((4.10) + (2.40) + (3.20) + (1.50)) = 230$



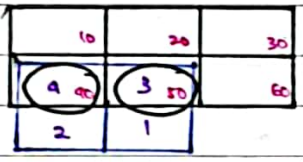
$Y(2,1) = ((4.20) + (2.50) + (3.50) + (1.60)) = 330$



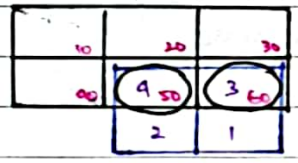
$Y(3,1) = ((4.30) + (2.60)) = 240$



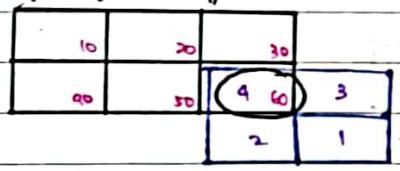
$Y(1,2) = ((4.40) + (3.50)) = 310$



$Y(2,2) = ((4.50) + (3.60)) = 380$



$Y(3,2) = (4.60) = 240$



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② konvolusi 2D secara matematis

persamaan :

$$Y[m,n] = \sum_{i=0}^{I-1} \sum_{j=0}^{J-1} g[i,j] \cdot F[m-i, n-j]$$

	0	1	2	3
0	10	40	70	60
1	70	230	330	240
2	120	310	380	240

M

N

$M = X + I - 1$ → banyaknya kolom

$N = Y + J - 1$ → banyaknya baris

* contoh soal

$g[i,j] = [\overset{0,0}{10}, \overset{1,0}{20}, \overset{2,0}{30} ; \overset{0,1}{90}, \overset{1,1}{50}, \overset{2,1}{60}]$ $I = 3$ dan $J = 2$

$F[x,y] = [\overset{0,0}{1}, \overset{1,0}{2} ; \overset{0,1}{3}, \overset{1,1}{4}]$ $x = 2$ dan $Y = 2$

Dit : tentukan hasil konvolusinya!

— jawab —

$M = X + I - 1$ $N = Y + J - 1$

$= 2 + 3 - 1$ $= 2 + 2 - 1$

$M = 4$ $N = 3$

$Y[0,0] = g[0,0] \cdot F(0-0, 0-0) + g[0,1] \cdot F[0-0, 0-1] +$
$g[1,0] \cdot F$
$g[2,0]$

$Y[0,0] = g[0,0] \cdot F[0,0]$

$Y[0,0] = 10.1 = 10$ //

$Y[1,0] = g[0,0] \cdot F[1-0, 0-0] + g[0,1] \cdot F[1-0, 0-1] +$
$g[1,0] \cdot F[1-1, 0-0] + g[1,1] \cdot F[1-0, 0-1] +$
$g[2,0] \cdot F[1-2, 0-0] + g[2,1] \cdot F[1-2, 0-1]$

$Y[1,0] = g[0,0] \cdot F[1,0] + g[1,0] \cdot F[0,0]$

$= (10 \cdot 2) + (20 \cdot 1)$

$Y[1,0] = 40$ //

$Y[2,0] = g[0,0] \cdot F[2-0, 0-0] + g[0,1] \cdot F[2-0, 0-1] +$
$g[1,0] \cdot F[2-1, 0-0] + g[1,1] \cdot F[2-1, 0-1] +$
$g[2,0] \cdot F[2-2, 0-0] + g[2,1] \cdot F[2-2, 0-1]$

$= g[1,0] \cdot F[1,0] + g[2,0] \cdot F[0,0]$

$= (20 \cdot 2) + (30 \cdot 1)$

$Y[2,0] = 70$ //

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Y[2,1] = g[0,0].F[2-0,1-0] + g[0,1].F[2-0,1-1] +
g[1,0].F[2-1,1-0] + g[1,1].F[2-1,1-1] +
g[2,0].F[2-2,1-0] + g[2,1].F[2-2,1-1]

Y[2,1] = g[0,0].F[2,1] + g[0,1].F[2,0] + g[1,0].F[1,1] + g[1,1].F[1,0] +
g[2,0].F[0,1] + g[2,1].F[0,0]

= (20.9) + (10.2) + (30.3) + (60.1)

= 80 + 100 + 90 + 60

Y[2,1] = 330 //

Y[3,0] = g[0,0].F[3-0,0-0] + g[0,1].F[3-0,3-1] +
g[1,0].F[3-1,0-0] + g[1,1].F[3-1,0-1] +
g[2,0].F[3-2,0-0] + g[2,1].F[3-2,0-1]

Y[3,0] = g[0,0].F[3,0] + g[0,1].F[3,2] + g[1,0].F[2,0] + g[2,0].F[1,0]

= (30.2) = 60 //

Y[0,1] = g[0,0].F[0-0,1-0] + g[0,1].F[0-0,1-1] +
g[1,0].F[0-1,1-0] + g[1,1].F[0-1,1-1] +
g[2,0].F[0-2,1-0] + g[2,1].F[0-2,1-1]

Y[0,1] = g[0,0].F[0,1] + g[0,1].F[0,0]

= (10.3) + (40.1)

= 70 //

Y[1,1] = g[0,0].F[1-0,1-0] + g[0,1].F[1-0,1-1] +
g[1,0].F[1-1,1-0] + g[1,1].F[1-1,1-1] +
g[2,0].F[1-2,1-0] + g[2,1].F[1-2,1-1]

Y[1,1] = g[0,0].F[1,1] + g[0,1].F[1,0] + g[1,0].F[0,1] + g[1,1].F[0,0]

= (10.4) + (40.2) + (20.3) + (50.1)

= 230 //

Y[3,1] = g[0,0].F[3-0,1-0] + g[0,1].F[3-0,1-1] +
g[1,0].F[3-1,1-0] + g[1,1].F[3-1,1-1] +
g[2,0].F[3-2,1-0] + g[2,1].F[3-2,1-1]

Y[3,1] = g[2,0].F[1,1] + g[2,1].F[1,0]

= (30.4) + (60.2)

= 240 //

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$Y[0,2] = g[0,0].F[0-0, 2-0] + g[0,1].F[0-0, 2-1] +$
$g[1,0].F[0-1, 2-0] + g[1,1].F[0-1, 2-1] +$
$g[2,0].F[0-2, 2-0] + g[2,1].F[0-2, 2-1]$

$Y[0,2] = g[0,1].F[0,1]$
$= (40.3)$
$Y[0,2] = 120$ //

$Y[1,2] = g[0,0].F[1-0, 2-0] + g[0,1].F[1-0, 2-1] +$
$g[1,0].F[1-1, 2-0] + g[1,1].F[1-1, 2-1] +$
$g[2,0].F[1-2, 2-0] + g[2,1].F[1-2, 2-1].$

$Y[1,2] = g[0,1].F[1,1] + g[1,1].F[0,1]$
$= (40.4) + (50.3)$
$Y[1,2] = 310$ //

$Y[2,2] = g[0,0].F[2-0, 2-0] + g[0,1].F[2-0, 2-1] +$
$g[1,0].F[2-1, 2-0] + g[1,1].F[2-1, 2-1] +$
$g[2,0].F[2-2, 2-0] + g[2,1].F[2-2, 2-1]$

$Y[2,2] = g[1,1].F[1,1] + g[2,1].F[0,1]$
$= (50.4) + (60.3)$
$= 380$ //

$Y[3,2] = g[0,0].F[3-0, 2-0] + g[0,1].F[3-0, 2-1] +$
$g[1,0].F[3-1, 2-0] + g[1,1].F[3-1, 2-1] +$
$g[2,0].F[3-2, 0-2] + g[2,1].F[3-2, 2-1]$

$Y[3,2] = g[2,1].F[1,1]$
$= (60.4)$
$= 240$ //

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